

Cosmology Equations for Locking Report

1. Distances & Distance Modulus

$$\chi(z) = \int_0^z [c(z') / H(z')] dz'$$

$$S_k(\chi) = \{ \sin(\sqrt{|\Omega_k|} \chi) / \sqrt{|\Omega_k|} \text{ if } \Omega_k < 0; \chi \text{ if } \Omega_k = 0; \sinh(\sqrt{|\Omega_k|} \chi) / \sqrt{|\Omega_k|} \text{ if } \Omega_k > 0 \}$$

$$D_M(z) = S_k(\chi)$$

$$D_H(z) = c(z) / H(z)$$

$$D_A(z) = D_M(z) / (1 + z)$$

$$D_L(z) = (1 + z) \cdot D_M(z)$$

$$\mu(z) = 5 \log_{10} [D_L(z) / 10 \text{ pc}]$$

$$\Delta\mu(z) = \mu_{\text{obs}}(z) - \mu_{\text{model}}(z)$$

2. BAO Core Definitions

$$r_d = \int_{z_d}^{\infty} [c_s(z) / H(z)] dz$$

$$c_s(z) = c(z) / \sqrt{3(1 + R_b(z))}$$

$$R_b(z) = 3 \rho_b(z) / (4 \rho_\gamma(z))$$

$$\text{BAO observables: } D_M/r_d, D_H/r_d$$

$$D_V(z) = [z \cdot D_H(z) \cdot D_M^2(z)]^{1/3}$$

$$F_{AP}(z) = D_M(z) \cdot H(z) / c(z)$$

3. Expansion History

$$E(z) = H(z) / H_0 = \sqrt{[\Omega_r (1+z)^4 + \Omega_m(z)(1+z)^3 + \Omega_k(1+z)^2 + \Omega_{DE}(z) + \dots]}$$

$$\Omega_m(z) = \Omega_{m0} \cdot \mathcal{M}(z)$$

$$c(z) = c_0 \cdot \mathcal{C}(z)$$

4. Growth of Structure

$$f(z) = d \ln D_+(z) / d \ln a$$

$$f\sigma_8(z) = f(z) \cdot \sigma_8(z)$$

$$f(z) \approx [\Omega_m(z)]^\gamma \quad (\gamma \approx 0.55 \text{ in GR} + \Lambda\text{CDM})$$

5. CMB Distance Priors

$$R = \sqrt{[\Omega_m H_0^2]} \cdot D_M(z_*) / c(z_*)$$

$$\ell_a = \pi \cdot D_M(z_*) / r_s(z_*)$$

6. Locking & Statistics

$$r(\bar{\omega}, 1/k_{\text{eff}}) = \text{Cov}(\bar{\omega}, 1/k_{\text{eff}}) / (\sigma_{\bar{\omega}} \cdot \sigma_{\{1/k\}})$$

$$t = r \sqrt{[(N-2)/(1-r^2)]}$$

$$\chi^2 = (d - m)^T C^{-1} (d - m)$$

$$\text{AIC} = 2k + \chi^2_{\text{min}}$$

$$\text{BIC} = k \ln N + \chi^2_{\text{min}}$$

7. BAO/k-Space Link

$$k_{\text{BAO}}(z) \approx \pi / r_d$$

$$\lambda_{\text{BAO}}(z) \approx 2\pi / k_{\text{BAO}} \approx 2 r_d$$

$$\bar{\omega}(\log(1+z)) \leftrightarrow 1/k_{\text{eff}}$$